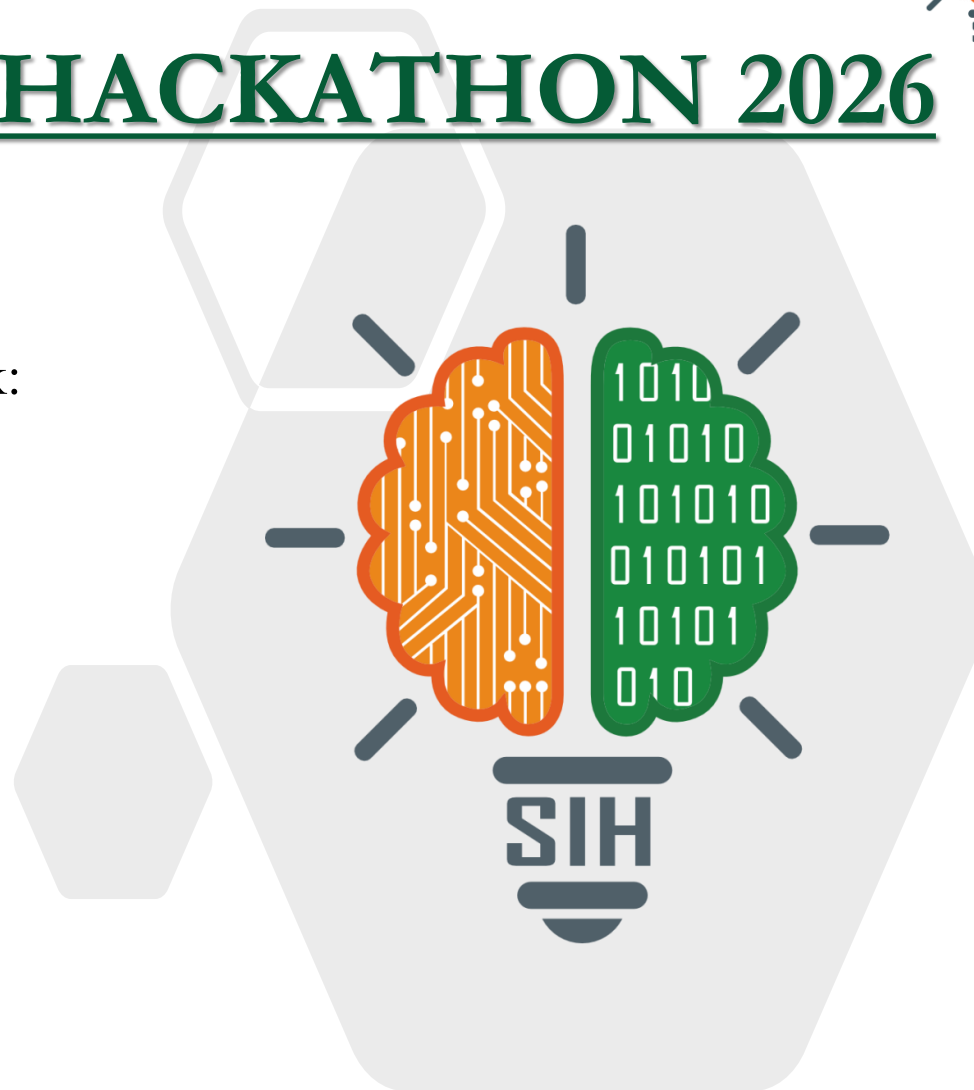


SMART INDIA HACKATHON 2026

- ♣ **Problem Statement ID** – 26045
- ♣ **Problem Statement Title-** IP-SAKTI Sahayak:
AI-Powered IP & Regulatory Navigator for
Ayurvedic Innovation
- ♣ **Theme-** MedTech / BioTech / HealthTech
(Ministry of AYUSH & AIIA)
- ♣ **PS Category-** Software
- ♣ **Team ID-** [Add your Team ID]
- ♣ **Team Name:** AyuRaksha



Proposed Solution: AyuRaksha

A multilingual, AI-powered assistant that helps Ayurveda innovators understand Intellectual Property, regulatory classification and biodiversity compliance through trusted, source-cited guidance. It supports Vaidyas, researchers, startups, MSMEs and cultivators in identifying the right protection, approval pathway and next steps for Ayurvedic products.

Problem Assessment

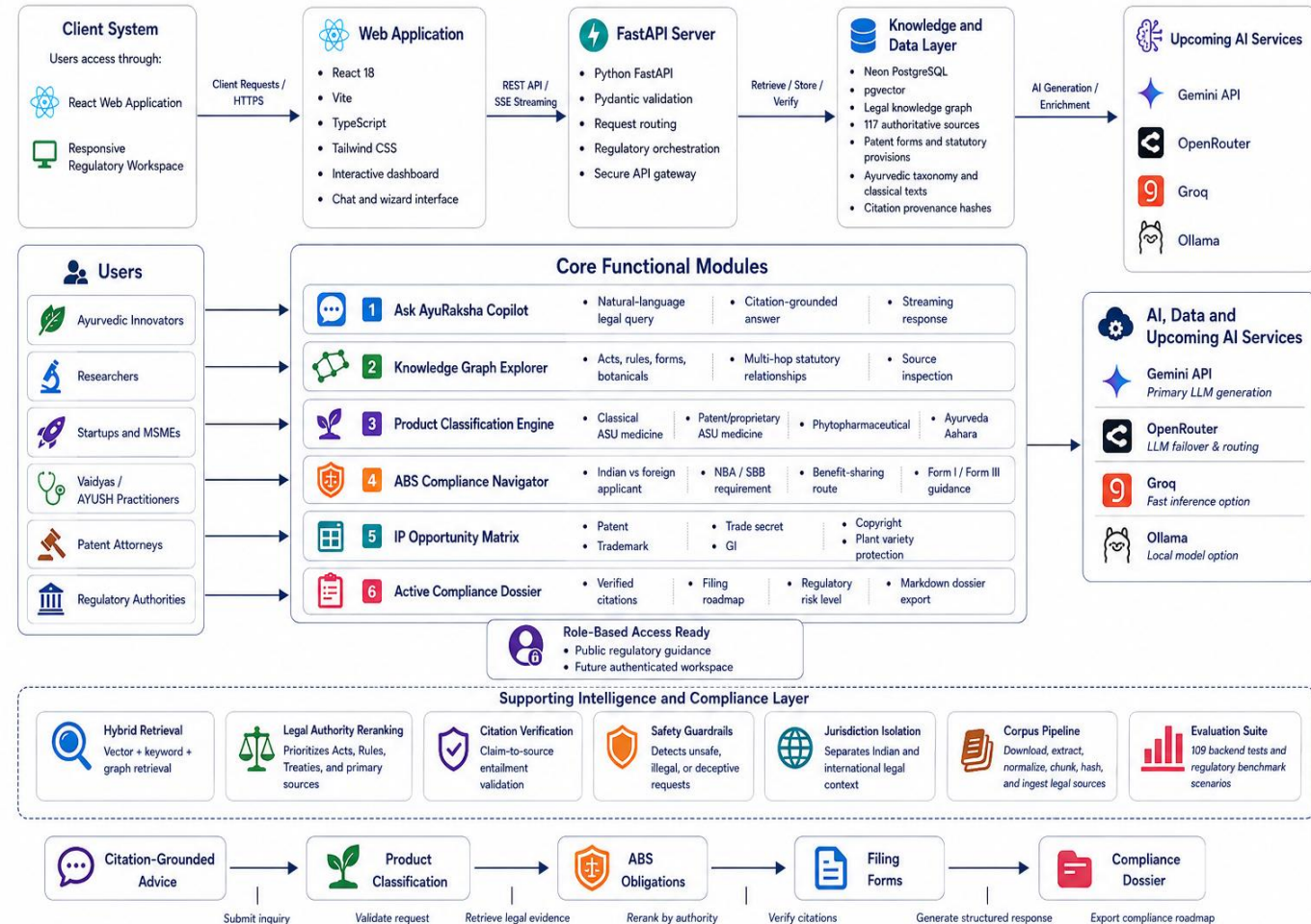
- ✓ Classifies the product as Classical Ayurvedic Medicine, Patent/ Proprietary Medicine or Cosmetic.
- ✓ Identifies the suitable IP route: Patent, Trade Mark, GI, Design, Copyright, Trade Secret or Plant Variety Protection.
- ✓ Flags Traditional Knowledge and prior-art risks, helping prevent patenting of known Ayurvedic formulations.
- ✓ Guides users on ABS compliance when Indian biological resources or traditional knowledge are involved.
- ✓ Provides a clear checklist for regulatory, labelling, advertising related next steps.

Innovation & Uniqueness

- ✓ Citation-grounded RAG: every important answer is linked to official laws, rules, treaties or government records.
- ✓ India and International modes remain separate, preventing confusion between domestic law and global IP requirements.
- ✓ Smart question flow asks only the minimum details needed to classify a formulation correctly.
- ✓ Confidence score, safe abstention and human-expert escalation for uncertain or high-risk cases.

AYURAKSHA — SYSTEM ARCHITECTURE

AI-Powered IP & Regulatory Navigator for Ayurvedic Innovation

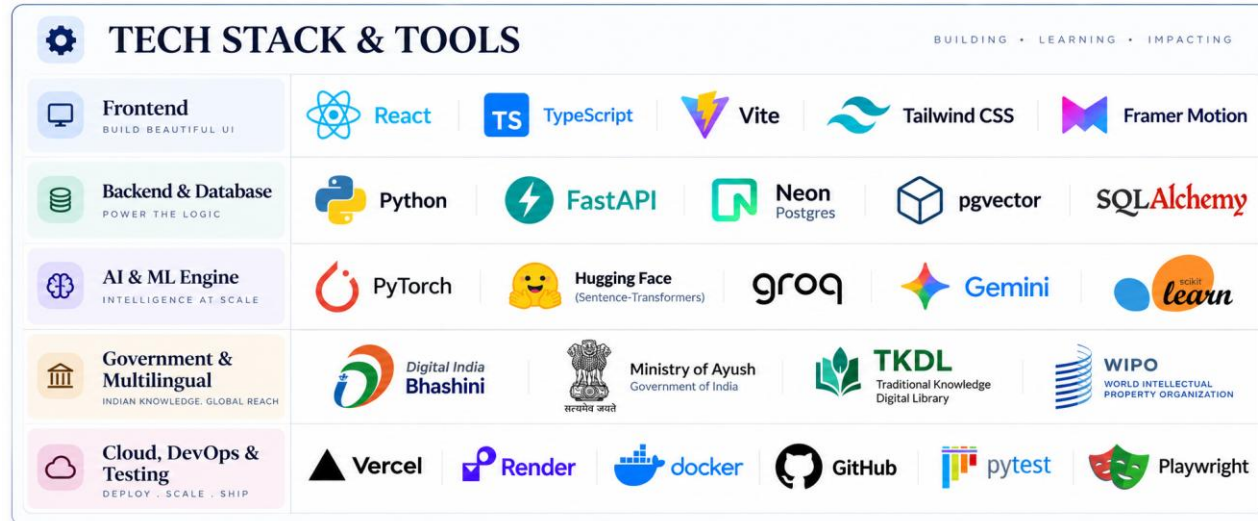


Technologies Used

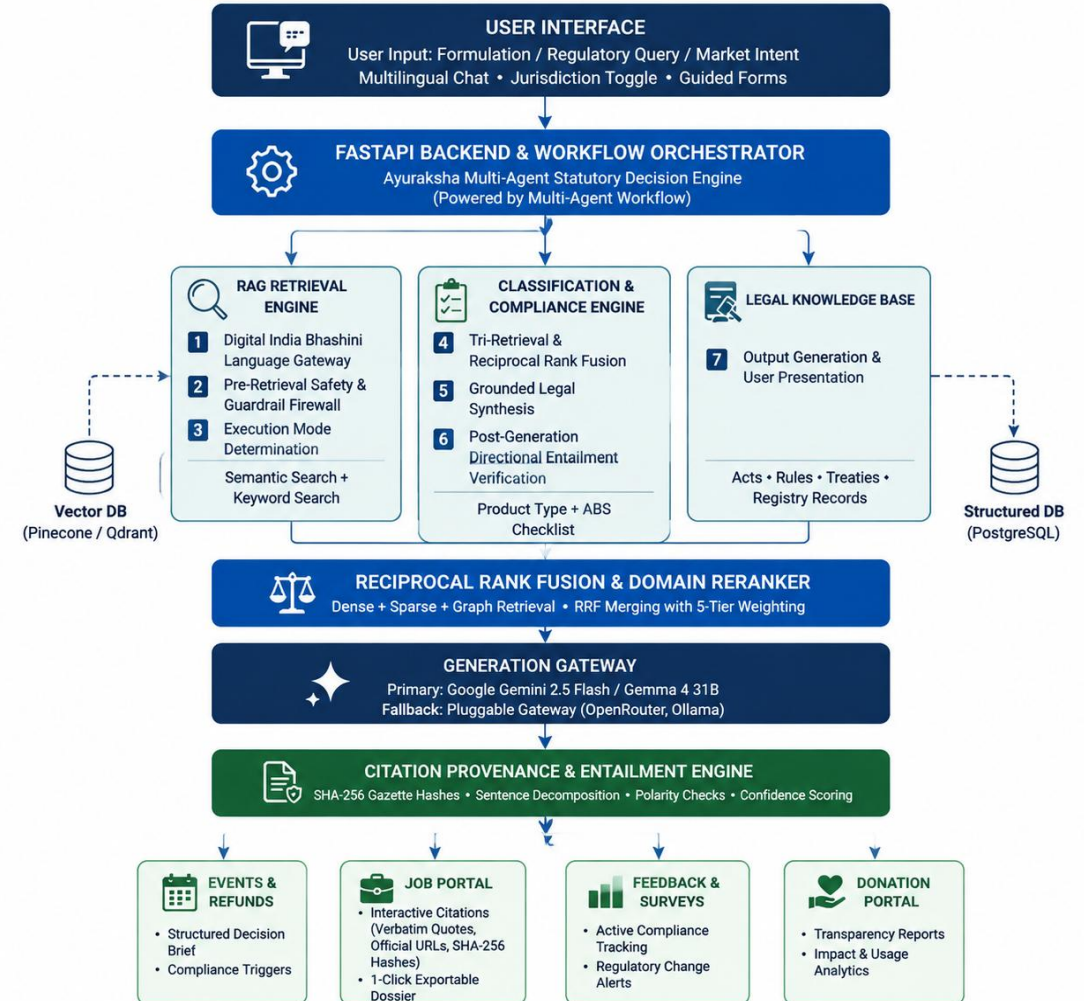
- Frontend: React / Streamlit-based multilingual chat interface with jurisdiction selector and guided question forms.
- Backend: Python + FastAPI for APIs, workflow orchestration and secure request handling.
- AI Layer: Large Language Model with Retrieval-Augmented Generation (RAG) for grounded legal answers.
- Language Support: Bhashini-compatible translation and speech layer for Indian-language interaction.
- Security: Role-based access, encrypted data handling, consent logging and audit trails.

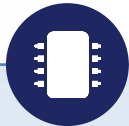
Methodology & Process for Implementation

- Collect official sources: India Code, IP India, NBA, TKDL, FSSAI, WIPO and other authorised public portals.
- Convert documents into structured, version-tagged text chunks with source title, section/article number, link and effective date.
- Retrieve relevant passages using semantic search and keyword-based legal search.
- Run a formulation-classification flow to determine the product category and applicable regulatory route.



SYSTEM ARCHITECTURE & FLOW





Technical Feasibility

- Built using proven technologies: web interface, Python backend, RAG pipeline and vector database.
- Modular architecture enables gradual addition of new laws, languages, sources and compliance workflows.



Operational Feasibility

- Guided questions simplify complex IP and regulatory decisions for Vaidyas, startups, researchers and MSMEs.
- Official citations, source links and human escalation make the guidance practical and trustworthy.



Financial Feasibility

- The MVP can use open-source tools and freely available official sources, keeping initial costs low.
- Phased rollout enables affordable scaling from citation-RAG to multilingual voice and advanced integrations.

 *MVP success will be evaluated using accuracy, citation quality, safe abstention and user-task completion metrics.*



0.00%

High Citation Coverage



100%

Verified Official Sources



100%

Clear India/International Separation



100%

Safe Abstention for Uncertain Queries



100%

Multilingual User Accessibility



POTENTIAL CHALLENGES & RISKS



Laws, rules and notifications may change, making older information unreliable.



Incomplete user details can lead to an incorrect preliminary product classification.



Users may treat AI guidance as final legal advice without expert verification.



Product formulas, sourcing information and business details require strong privacy protection.



STRATEGIES FOR OVERCOMING THEM



Maintain a version-controlled corpus with regular updates from official sources.



Ask focused clarifying questions before generating product-specific guidance.



Display confidence levels, clear disclaimers and human-expert escalation for complex cases.



Use consent-based access, encryption, minimal data collection and secure audit logs.

Jurisdictional Clarity:

- Impact: Separates India-specific guidance from international IP and treaty requirements.
- Benefit: Prevents users from confusing Indian compliance steps with export-market or global filing requirements.

Faster, Safer Formulation Classification:

- Impact: Helps identify whether a product is a classical medicine, proprietary medicine, food, cosmetic or phytopharmaceutical.
- Benefit: Users understand the right regulatory and IP pathway before investing in filing or launch activities.

Reduced Misappropriation Risk:

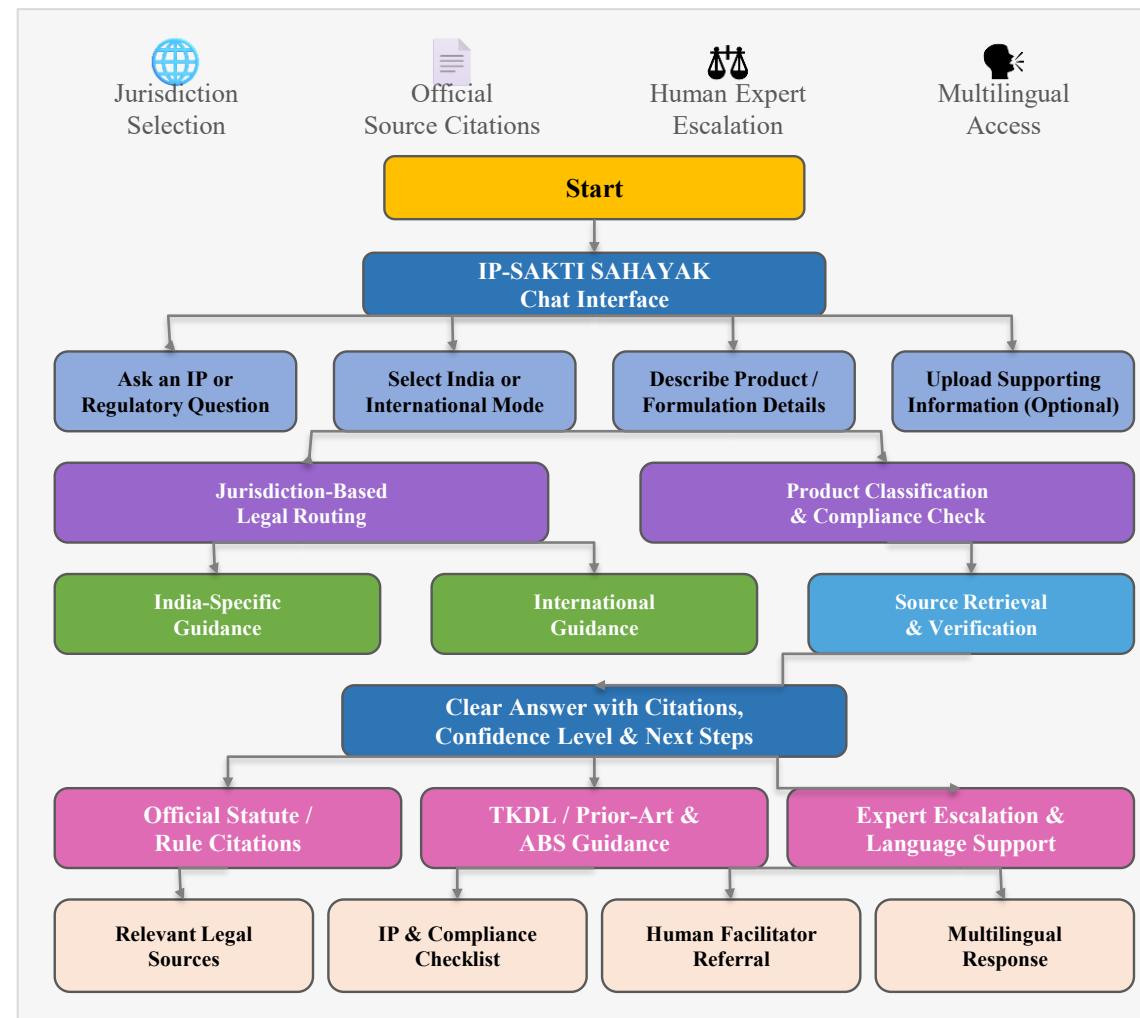
- Impact: Highlights traditional-knowledge and prior-art concerns in Ayurvedic formulations.
- Benefit: Helps reduce misappropriation risk and directs users to TKDL/prior-art verification resources.

Evidence-Backed, Trustworthy Answers:

- Impact: Answers are generated from curated official sources and displayed with citations and confidence levels.
- Benefit: Users receive more transparent, verifiable guidance and can escalate complex cases to an expert.

Inclusive, Scalable Reach:

- Impact: Multilingual delivery makes IP and regulatory guidance easier to understand for diverse users.
- Benefit: Supports Vaidyas, cultivators, researchers, startups and MSMEs across India.



Research And References

- **India Code — Patents Act, 1970 and Patents Rules: Patentability, traditional-knowledge exclusions and patent filing procedures.**
- **India Code — Biological Diversity Act, 2002 and related Rules: Access and Benefit Sharing (ABS) obligations for biological resources.**
- **Drugs and Cosmetics Act, 1940 and Rules, 1945: Regulatory classification and requirements for Ayurvedic, Siddha and Unani medicines.**
- **FSSAI — Food Safety and Standards regulations, including Ayurveda-Aahar requirements: Food-category, labelling and claim-related guidance.**
- **IP India — Public databases and official information for patents, trade marks, designs and geographical indications.**
- **Traditional Knowledge Digital Library (TKDL): Prior-art resource supporting protection of India's traditional medicinal knowledge.**
- **National Biodiversity Authority (NBA): Official ABS guidance, approvals, notifications and biodiversity-related compliance resources.**
- **WIPO — PCT, Madrid System, Hague System, Nagoya Protocol and WIPO GRATK Treaty resources for international IP awareness.**

Our knowledge base is built from authoritative public sources and stores document title, jurisdiction, section/article reference, source link and update date. Every response cites the retrieved source, shows a confidence level and recommends expert review where the law or facts are uncertain.